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 # Network Protocols: DNS, HTTP/HTTPS, and Packet Analysis

Tasks

1. Use Wireshark to capture network traffic while you load www.spotify.com in your browser, then identify and note down at least two DNS packets and two HTTP or HTTPS packets in the capture.

Answer:

Steps:

1. Open **Wireshark**

2. Select your active WiFi network

3. Click **Start Capture**

4. Open browser and type:
www.spotify.com

5. Wait for the page to load fully

6. Stop the capture in Wireshark

7. In the filter bar, type:

```text id="dns01"
dns
```

and then:

```text id="https01"

http or https

```

DNS Packets (Examples you may see):

** **Standard query:** www.spotify.com*

** This is your computer asking for Spotify's IP address*

* **Standard query response:** www.spotify.com → 104.xxx.xxx.xxx

* This is DNS server replying with Spotify’s IP address

HTTP/HTTPS Packets (Examples you may see):

* **Client Hello (HTTPS):**

* Your browser starts a secure connection with Spotify

* **Server Hello / Application Data (HTTPS):**

* Spotify server responds and sends encrypted website data

Explanation:

DNS packets are used to convert website names into IP addresses, while HTTPS packets are used to securely load website content like pages, images, and login data.

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2. Simulate a DNS lookup for www.flipkart.com using the nslookup command in your terminal and write down

the IP address returned along with a brief explanation of what this result means.

Answer:

Command:

```cmd id="nslookup01"
nslookup www.flipkart.com
```

Example Output:

```text id="nslookup02"
Name: www.flipkart.com
Address: 163.xxx.xxx.xxx
```

```

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### Explanation:

This result shows the **IP address of Flipkart’s server**. DNS converts the domain name ([www.flipkart.com](http://www.flipkart.com)) into this IP address so your browser can connect to the correct server on the internet.

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### 3. Explain the difference between HTTP and HTTPS by listing 3 key differences, and give a real-world example of an app or website you use that relies on HTTPS for secure communication.
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Answer:

1. Security:

* HTTP sends data in plain text (not secure)
* HTTPS encrypts data for security

2. Data Protection:

* HTTP data can be stolen or modified
* HTTPS protects data using encryption (SSL/TLS)

3. Usage:

* HTTP is used for basic or older websites

* HTTPS is used for secure websites like banking, shopping, and social media

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Real-World Example:

* **Instagram** uses HTTPS to protect login details, messages, and personal data
* **Paytm** also uses HTTPS for secure financial transactions

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4. Open Wireshark and filter for DNS protocol traffic. Try searching for a new website (one you haven't visited recently) in your browser and describe

the sequence of DNS queries and responses you observe.

Answer:

Steps:

1. Open Wireshark
2. Start capture
3. In filter bar type:

```text id="dnsfilter01"
dns
```

4. Search a new website in browser (example: a news or shopping site)

5. Observe packets

Sequence of DNS Activity:

*1. **Standard Query (Request):***

** Your device asks DNS server for IP address of the website*

*2. **Query Sent to DNS Server:***

** The request is forwarded to ISP or Google DNS*

3. **Standard Query Response:**

* DNS server replies with IP address of the website

4. **Browser Connects to Server:**

* After receiving IP, browser starts loading the website

Explanation:

DNS works like a phonebook of the internet. It first finds the correct IP address before your browser can open any website.

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5. Pick any one protocol from DNS, DHCP, FTP, or SNMP (other than HTTP/HTTPS) and write a short scenario describing how it would be used in a real-world app or service you use.

Answer:

Protocol Chosen: DHCP (Dynamic Host Configuration Protocol)

Real-World Scenario:

When you go to a café and connect your phone to their WiFi to open Instagram, the **DHCP server *automatically assigns your phone an IP address** like 192.168.1.20.*

Without DHCP, you would have to manually enter IP settings every time you connect to a new network.

Explanation:

DHCP makes connecting to WiFi automatic and easy by giving devices IP addresses, subnet mask, and default gateway instantly. This allows apps like Instagram, YouTube, and WhatsApp to work without manual network setup.